An elementary embedding of a free system into itself is surjective

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Editorial note. The counted statement is the author's environment at source0.tex lines 1076-1077 and its proof environment opens at line 1078; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial scope. One original result of the pinned author manuscript, counted once: elementary embeddings of the free system into itself are surjective (author label April_11_2024_5).

Allowed standard prerequisites (editorial).

- Free system on finitely many generators. F_n is freely generated by a set of n elements, so its generated subsystems correspond to subsets of the free base.. It is the object of the statement.
- Elementary embedding. An embedding preserving the truth of all formulas with parameters.. It is the hypothesis.
- Surjective endomorphism is an automorphism. Every surjective homomorphism of F_n onto itself is an automorphism.. It converts surjectivity into the automorphism conclusion.

Remark 0.1 Let F_n be the free STS with $n \geq 3$ generators. Every surjective homomorphism $f : F_n \to F_n$ is an automorphism of F_n.

Proposition 0.2 Let F_n be the free STS with $n \geq 3$ generators. Every elementary embedding $f : F_n \to F_n$ is surjective and, therefore, is an automorphism of F_n.

Proof. Let $F_n = \langle \bar{a} \rangle$, where $\bar{a} = a_1, \ldots, a_n$ is a free base, and let $a'_i = f(a_i) = t_i(\bar{a})$ for every $i = 1, \ldots, n$, where t_i is a corresponding term. Then F_n satisfies

$$\exists x_1 \ldots x_n \bigwedge_{i=1}^{n} a'_i = t_i(x_1, \ldots, x_n).$$

Since $f(F_n) \preceq F_n$, the same sentence holds in $f(F_n)$ and, by isomorphism, F_n satisfies

$$\exists x_1 \ldots x_n \bigwedge_{i=1}^{n} a_i = t_i(x_1, \ldots, x_n).$$

Choose $\bar{b} = b_1, \ldots, b_n \in F_n$ such that $a_i = t_i(\bar{b})$ for every $i = 1, \ldots, n$. Then $F_n = \langle b_1, \ldots, b_n \rangle$ and $b_1, \ldots, b_n$ are independent by Remark 0.1. Now consider the automorphism $g : F_n \to F_n$ defined by $g(b_i) = a_i$ for $i = 1, \ldots, n$. Observe that for every i

$$g(a_i) = g(t_i(\bar{b})) = t_i(g(\bar{b})) = t_i(\bar{a}) = a'_i.$$

This implies that $f = g$ and, therefore, that f is an automorphism. □